

Nathaniel Mapaye

Software Engineer

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Summary

Software engineer and Technology & Information Management student at UC Santa Cruz building reliable systems from embedded FreeRTOS and concurrent C++ to React Native mobile products. Experience spans AI evaluation, data-center engineering, cybersecurity, and technical tutoring.

Experience

AI Fellow - Handshake

Apr 2026 - Present
Remote / Contract

- Evaluate paired AI responses for correctness, completeness, clarity, and usefulness; document concrete strengths, failure modes, and edge cases.
- Review repository behavior, diffs, tests, and implementation logic to assess whether proposed software changes satisfy task requirements.

Linear Algebra Tutor - UCSC ACE

Jan 2026 - Apr 2026
Santa Cruz, CA

- Led guided problem-solving sessions covering systems of equations, vector spaces, linear transformations, eigenvalues, and diagonalization.
- Adapted explanations across skill levels and helped students move from memorized procedures to step-by-step reasoning.

Data Center Engineering Intern - Bitera D.C

Aug 2025 - Sep 2025
Jakarta, Indonesia

- Coordinated remediation and verification for high-severity web vulnerabilities and documented repeatable validation steps.
- Authored incident postmortems and operations runbooks covering power, cooling, network, cabling, and alarm checks.

Cyber Security Analyst Intern - Xapiens Teknologi Indonesia

Jul 2024 - Sep 2024
Tangerang, Indonesia

- Performed authenticated scanning and manual validation, prioritized findings by exploitability and CVSS, and tracked remediation.
- Produced proof-of-concept-backed reports with concrete fixes and retest results.

Selected Projects

AURORA (React Native, TypeScript, HealthKit, MMKV) - iPhone and iPad app combining manual caffeine logs, optional read-only Apple Health sleep import, a 60-second vigilance test, and private on-device insights.

EmbNode (C++23, FreeRTOS, ESP32/STM32, MQTT) - DMA sampling pipeline with sampler, aggregation, communications, OTA, watchdog, and deep-sleep scheduling, plus host-simulation checks.

GitOps Canary Operator (Go, Kubernetes, Prometheus, ArgoCD) - SLO-driven progressive delivery controller with policy checks and automatic rollback when latency or error thresholds regress.

SysLib (C++23, atomics, lock-free data structures) - Header-only systems library with RAII utilities, SPSC and MPMC queues, sanitizer checks, benchmarks, and linearizability tests.

Education

B.S., Technology & Information Management - University of California, Santa Cruz

2023 - 2027
Santa Cruz, CA

- Undergraduate Dean's Scholarship Award; Bilingual International Baccalaureate Diploma.

Technical Skills

Languages: C++23, Go, TypeScript/JavaScript, Python, Java, Lua

Systems & mobile: FreeRTOS, ESP32/STM32, React Native, HealthKit, MMKV, Linux

Infrastructure & testing: Docker, Kubernetes, ArgoCD, Helm, Prometheus, GitHub Actions, CMake/CTest, Burp Suite, Nessus